# Predicting Cardiovascular Disease Risk via Supervised Learning

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***Abstract*—Predicting cardiovascular disease from simple features is an important challenge to improve infarction detection and prevention. To address the binary prediction of coronary heart disease / myocardial infarction (MICHD), we focused on the BRFSS 2015 survey dataset (328,135 observations, 321 raw features). With only 8.8% positive prevalence, F1-score provides a more appropriate metric than accuracy. Our best model, a weighted logistic regression regularized and Newton's method-optimized, reaches an F1-score of 0.433 on a 5-fold cross-validation and a test F1-score of 0.446 on AIcrowd. Our findings show the importance of features selection and threshold in the prediction quality, over the choice of learning model.**

## I. Introduction

Predicting myocardial infarction and coronary heart disease (MICHD) from behavioral risk factors is a challenge in public health prevention. In this report, we analyze a MICHD corpus (2015 Behavioral Risk Factor Surveillance System (BRFSS) with 328,135 observations, 321 features, positive prevalence of 8.8%) and apply different Machine Learning models in order to predict if a patient is sick or not, based on several features. As an evaluation metric, we used the F1-score, as the corpus is clearly imbalanced. Indeed, predicting *no disease* to all would lead to 91.2% accuracy, while F1-score would approach zero.

Initial experiments went through linear baselines (MSE gradient descent, regularized logistic regression). They plateaued between 0.39 and 0.42, showing that performance limit does not come from the algorithms themselves. Our work shows:

- Extraction of signal from missing values and selection of variables;
- A leak-free protocol and optimization of decision threshold;
- A numerical stabilization: elimination of colinear features for least squares and exact convergence with Newton method.

## II. Models and Methods

### A. Data Preprocessing and Feature Engineering

**Data cleaning**: Some survey intern codes (7, 9, 77, 99,...), used for example to indicate that people didn't respond to a question (refused/do not know), were converted to NaN values. We preserved `EMPLOY1` = 7 (which means "retired") and the code 88 was converted into 0 days. In the categorical features, "missing value" was considered as a category itself. Numeric variables undergo median imputation, an indication flag when missingness exceeds 5%, 1% and 99% percentile clipping to suppress outliers, and standardization (z-score). Constant columns and categories with less than 20 rows were discarded.

Code and reproduction scripts available at https://github.com/MohamedZirar5/ML_Project1

Initial data analysis reveals key challenges from the dataset: categorical features, severe class imbalance (91.2% for class $-1$ versus 8.8% for class 1), and labels in $\{-1, 1\}$ (instead of $\{0, 1\}$ required for standard logistic regression). This heavy imbalance renders accuracy an uninformative metric, as a trivial model predicting only the majority class would achieve high accuracy.

**Feature selection**: Lots of features don't have any answer, but it does not come randomly, it expresses conditional reasons. For instance, BPMEDS (medicine against hypertension) is asked only if hypertension has been proved. Thus, 60% of patients lack this information (with 3.8% of disease cases), whereas 16.3% of respondents who answered are concerned by the infarction. Out of 321 raw columns, we retain 228: the 45 baseline columns plus three restored groups (*redundant* +0.0067, *risk_management* +0.0025 and *optional_modules* +0.0019). We excluded `cvd_care` (post-event rehabilitation and aspirin use), as it constitutes label leakage. We expanded feature space with standardized squares of numeric columns and pairwise interactions with age `_AGE80` and `SEX=2`, yielding to 1236 columns.

### B. Models

We used **6 different standard models** :

- Least squares;
- Gradient Descent (GD) with MSE;
- Stochastic Gradient Descent with MSE;
- Ridge Regression (with a penalty $\frac{\lambda}{2}\|\mathbf{w}\|_2^2$);
- Logistic Regression (without and with regularization).

The MSE loss incorporates a $\frac{1}{2}$ factor, and reported losses exclude the regularization penalty.

**Two optimized methods**

- Analytical least squares;
- Optimized logistic regression with Newton method.

As the feature covariance matrix is ill-conditioned ($X^TX$ over 1236 features is singular), we removed 85 linearly dependent columns to ensure an invertible solution. Besides that, first-order GD requires a tiny step size ($\gamma < \frac{8}{L}$) and does not reach the optimum even after 600 iterations. On the contrary, Newton's method reaches the exact minimum in only 8 iterations.

TABLE I
CROSS-VALIDATION AND TEST PERFORMANCE (MEAN ± STD OVER 5 FOLDS).

| Model | Hyperparameters | CV F1-Score | Test F1-Score |
|---|---|---|---|
| Baseline (Ridge) | 45 features, $\lambda = 10^{-2}$ | 0.412 ±0.0061 | – |
| Least squares | $D = 1151$ (85 linearly dependent features pruned) | 0.4306 ±0.0052 | – |
| Weighted least squares | positive weight = 3 | $0.4314 \pm 0.0038$ | 0.442 |
| Ridge regression | $\lambda = 10^{-3}$ | $0.4317 \pm 0.0045$ | 0.441 |
| Reg. logistic reg. (Newton) | $\lambda = 10^{-3}$, positive weight = 2 | $0.4326 \pm 0.0046$ | 0.446 |

### C. Validation Protocol

We used a 5-fold cross-validation, conserving the 8.83% ratio of positive cases in each fold and fixed a random seed (seed = 1).

**Training set**: All training operations have been made only on training set. As a leakage demonstration, oversampling prior fold split yields to an artificial F1-score of 0.7028, whereas correct in-fold balancing produces $F1 = 0.4206$.

**Threshold tuning**: The threshold 0.5 was not well-suited to asymmetric distributions ($F1 = 0.09$ with least squares). We optimized the threshold on out-of-fold predictions to maximize F1-score. For least squares method, it raises at 0.21, leading to $F1 \approx 0.41$ (Fig. 1). The optimal threshold for our best model is a linear score of $-0.6323$ (with probability 0.3470).

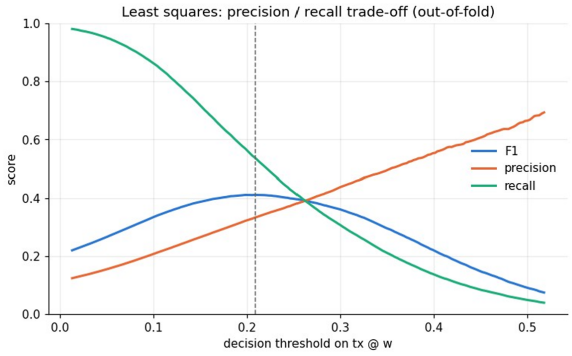



Fig. 1. Precision, recall, and $F1$-score trade-off on out-of-fold least squares predictions as a function of the decision threshold. The default 0.5 threshold yields an $F1$ of 0.09, whereas calibration at 0.210 attains 0.411.

## III. RESULTS

Table I compares our final model configurations on the full dataset (1,236 features) against the AIcrowd benchmark. The fold-to-fold standard deviation is 0.004 to 0.006 meaning that differences below 0.004 are not significant differences between results.

The step-by-step impact of features additions and rebalancing is presented in Table II. Starting from the baseline feature set ($F1 = 0.4111$), each stage of ablation and undersampling augmented the performance to 0.4317.

## IV. DISCUSSION

All linear algorithms (least squares, Ridge regression, Logistic regression) plateau around 0.43 of F1-score with cross-validation. This confirms our postulation that limitation is not present in the model, but in the features themselves. Testing other interactions (20 or 40 best predictors) do not improve the result (0.4322 and 0.4314). Random Fourier Features do not improve results neither (0.4328). A $1:1$ undersampling degrades performance on the full feature space ($F1 = 0.4278$), by discarding 90% of negative instances; a $1:5$ ratio ($F1 = 0.4326$) preserves majority-class variance, while penalizing false negative. Averaging 10 Ridge models (0.4296) or weighted results ($0.7 \times$ Logistic $+ 0.3 \times$ WLS $= 0.4328$) do not make a significant difference with main results.

TABLE II
FEATURE ABLATION STUDY (RIDGE REGRESSION, $\lambda = 10^{-3}$).

| Engineering Step | Features ($D$) | CV F1 (mean ± std) |
|---|---|---|
| Notebook 01 baseline | 178 | $0.4111 \pm 0.0066$ |
| + In-fold undersample 1:1 | 178 | $0.4213 \pm 0.0050$ |
| + 3 feature groups restored | 816 | $0.4255 \pm 0.0044$ |
| + Squares of numeric columns | 873 | $0.4258 \pm 0.0039$ |
| + Age and sex interactions | 1236 | $0.4278 \pm 0.0061$ |
| + Tuned ($\lambda = 10^{-3}$, undersample 1:5) | 1236 | $\mathbf{0.4317 \pm 0.0045}$ |

Linear SVMs were omitted as they yield an equivalent linear boundary, while kernel SVMs were computationally intractable on 328,135 observations without external libraries.

**Limitations**: Calibrating the decision threshold introduces a small bias, that affects all evaluated models equally. Moreover, AICrowd submissions have an intrinsic variance of 0.005, due to the sample test.

## V. SUMMARY

We demonstrated that extracting signal from survey, restoring informative feature groups, suppressing collinearities and optimizing decision threshold improved disease prediction, lifting the F1-score from 0.09 to 0.446.

**Reproducibility**: The `run.py` script is available on GitHub and runs in less than 3 minutes to reproduce the 0.446 AIcrowd result.